

Tab: READ ME

Data dictionary for GOOS EOVS: Seagrass Cover and Composition

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This data dictionary and schema provide definitions and units for variables used in reporting the Global Ocean Observing System's (GOOS) Essential Ocean Variable (EOV) 'Seagrass Cover and Composition', organized into the three tables required for submission of EOVS data to the Ocean Biodiversity Information System (OBIS): Event, Occurrence, and Extended-Measurement-or-Fact (EMoF) tables. The variables use a standard format compliant with Darwin Core vocabulary and include terms required by OBIS for sampling events, occurrences, and taxonomy, as well as a set of recommended terms specific to seagrass systems presented in the OBIS Extended Measurement-or-Fact (EMoF) module. Taxonomy of organisms is expected to follow standards of the World Register of Marine Species (WoRMS).

Prepared with support from the Scientific Committee for Oceanic Research (SCOR) to Working Group 158, "Coordinated Global Research Assessment of Seagrass Systems (C-GRASS)" and the Global Ocean Observing System (GOOS) Biology and Ecosystems panel Seagrass EOVS team

The data package required for submission of a seagrass EOVS dataset to OBIS consists of three csv files: event core, occurrence, and EMoF.

Tab	Explanation
READ ME	Explanation of the components of the data schema.
Terms and tables	List and definitions of variables in the three core tables (files) for a data package submitted to OBIS, including those required and recommended for the seagrass EOVS.
Extended Measurement-or-Fact	List and definitions of variables specific to the seagrass EOVS and included in the Extended Measurements or Facts (EMoF) table.
Example event	Example of empirical data for sampling events in seagrass habitats of the Indian River Lagoon, Florida, USA. The example is illustrative, containing only a subset of the data collected during these events (see OBIS: https://obis.org/dataset/beab1dcb-8235-4072-85cd-d8c3373e4ae0)
Example occurrence	Example of empirical data for occurrences of seagrass and associated organisms from the seagrass sampling events in the Indian River Lagoon, Florida, USA.
Example Extended Measurement-or-Fact	Example of empirical data for abundances of seagrass and associated organisms from the seagrass sampling events in the Indian River Lagoon, Florida, USA.

Tab: Terms and Tables

File (Table)	Dataset component (DwC archives)	DwC term	OBIS: required	Seagrass EOY: required	Seagrass: recommended	Definition (DwC)
File 1: Event	Event (Core Event table)	decimalLatitude	required	required	required	The geographic latitude (in decimal degrees, using the spatial reference system given in geodeticDatum) of the geographic center of a Location. Positive values are north of the Equator, negative values are south of it. Legal values lie between -90 and 90, inclusive.
File 1: Event	Event (Core Event table)	decimalLongitude	required	required	required	The geographic longitude (in decimal degrees, using the spatial reference system given in geodeticDatum) of the geographic center of a Location. Positive values are east of the Greenwich Meridian, negative values are west of it. Legal values lie between -180 and 180, inclusive.
File 1: Event	Event (Core Event table)	eventDate	required	required	required	The date-time or interval during which an Event occurred. For occurrences, this is the date-time when the event was recorded. Not suitable for a time in a geological context.
File 1: Event	Event (Core Event table)	eventID	required	required	required	An identifier for the set of information associated with an Event (something that occurs at a place and time). May be a global unique identifier or an identifier specific to the data set.
File 1: Event	Event (Core Event table)	bibliographicCitation			recommended	A bibliographic reference for the resource (e.g., journal article), if available.
File 1: Event	Event (Core Event table)	country			recommended	The name of the country or major administrative unit in which the Location occurs.
File 1: Event	Event (Core Event table)	datasetID			recommended	An identifier for the set of data. May be a global unique identifier or an identifier specific to a collection or institution.
File 1: Event	Event (Core Event table)	datasetName			recommended	The name identifying the data set from which the record was derived.
File 1: Event	Event (Core Event table)	fieldNotes			recommended	One of a) an indicator of the existence of, b) a reference to (publication, URI), or c) the text of notes taken in the field about the Event.
File 1: Event	Event (Core Event table)	habitat			recommended	A category or description of the habitat in which the Event occurred.
File 1: Event	Event (Core Event table)	institutionCode			recommended	The name (or acronym) in use by the institution having custody of the object(s) or information referred to in the record.
File 1: Event	Event (Core Event table)	license			recommended	A legal document giving official permission to do something with the resource.
File 1: Event	Event (Core Event table)	maximumDepthInMeters			recommended	The greater depth of a range of depth below the local surface, in meters.
File 1: Event	Event (Core Event table)	minimumDepthInMeters			recommended	The lesser depth of a range of depth below the local surface, in meters.
File 1: Event	Event (Core Event table)	parentEventID			recommended	An identifier for the broader Event that groups this and potentially other Events.
File 1: Event	Event (Core Event table)	rightsHolder			recommended	A person or organization owning or managing rights over the resource.
File 1: Event	Event (Core Event table)	samplingProtocol			recommended	The names of, references to, or descriptions of the methods or protocols used during an Event.
File 2: Occurrence	Occurrence (Occurrence table)	eventID	required	required	required	This term is the same as that in the event file and serves to link the occurrence data to the event data.
File 2: Occurrence	Occurrence (Occurrence table)	basisOfRecord	required	required	required	The specific nature of the data record (e.g., human observation, machine observation, fossil specimen).
File 2: Occurrence	Occurrence (Occurrence table)	occurrenceID	required	required	required	An identifier for the Occurrence (as opposed to a particular digital record of the occurrence). In the absence of a persistent global unique identifier, an occurrenceID may be constructed from a combination of identifiers in the record that will most closely make the occurrenceID globally unique.
File 2: Occurrence	Occurrence (Occurrence table)	occurrenceStatus	required	required	required	A statement about the presence or absence of a Taxon at a Location.
File 2: Occurrence	Occurrence (Occurrence table)	scientificName	required	required	required	The full scientific name, with authorship and date information if known. When forming part of an identification, this should be the name in lowest level taxonomic rank that can be determined. This term should not contain identification qualifications, which should instead be supplied in the IdentificationQualifier term.
File 2: Occurrence	Occurrence (Occurrence table)	scientificNameID	required	required	required	An identifier for the nomenclatural (not taxonomic) details of a scientific name.
File 2: Occurrence	Occurrence (Occurrence table)	IdentificationQualifier			recommended	A brief phrase or a standard term ("cf.", "aff.") to express the determiner's doubts about the dwc:Identification.
File 2: Occurrence	Occurrence (Occurrence table)	recordedBy			recommended	A list (concatenated and separated) of names of people, groups, or organizations who assigned the Taxon to the subject.
File 2: Occurrence	Occurrence (Occurrence table)	recordedBy			recommended	A list (concatenated and separated) of names of people, groups, or organizations responsible for recording the original Occurrence. The primary collector or observer, especially one who applies a personal identifier (recordNumber), should be listed first.
File 2: Occurrence	Occurrence (Occurrence table)	vernacularName			recommended	A common or vernacular name.
File 3: EMoF	Extended Measurement or Fact	eventID	required	required	required	This term is the same as that in the event file and serves to link the occurrence data to the event data.
File 3: EMoF	Extended Measurement or Fact	occurrenceID	required	required	required	This term is the same as that in the occurrence file and serves to link the extended measurement data to the occurrence data.
File 3: EMoF	Extended Measurement or Fact	measurementType	required	required	required	The nature of the measurement, fact, characteristic, or assertion.
File 3: EMoF	Extended Measurement or Fact	measurementUnit	required	required	required	The units associated with the measurementValue.
File 3: EMoF	Extended Measurement or Fact	measurementValue	required	required	required	The value of the measurement, fact, characteristic, or assertion.
File 3: EMoF	Extended Measurement or Fact	measurementDeterminedBy			recommended	A list (concatenated and separated) of names of people, groups, or organizations who determined the value of the MeasurementOrFact.
File 3: EMoF	Extended Measurement or Fact	measurementDeterminedDate			recommended	The date on which the MeasurementOrFact was made.
File 3: EMoF	Extended Measurement or Fact	measurementMethod			recommended	A description of or reference to (publication, URI) the method or protocol used to determine the measurement, fact, characteristic, or assertion.
File 3: EMoF	Extended Measurement or Fact	sampleSizeUnit			recommended	The unit of measurement of the size (time duration, length, area, or volume) of a sample in a sampling event.
File 3: EMoF	Extended Measurement or Fact	sampleSizeValue			recommended	A numeric value for a measurement of the size (time duration, length, area, or volume) of a sample in a sampling event.

Tab: Extended Measurement-or-Fact

Variable type	Variable name (EMoF)	measurementType (Definition of variable)	measurementUnit
Sub-variable (required)	percentCover	Proportion of substrate in a sample area that is covered (substrate not visible) by seagrass	Percent
Sub-variable (required)	arealExtent	Area in km2 occupied by seagrass meadow within spatial boundaries of a specified area	km ²
Supporting variable (biological)	canopyHeight	Median vertical distance from the sediment surface to the highest reach of seagrass leaves in the water column	cm
Supporting variable (biological)	shootDensity	The number of seagrass shoots per unit seabed area	Number m-2
Supporting variable (biological)	shootLength	Mean length of a random sample of seagrass shoots from seabed to tip of the leaf	cm
Supporting variable (biological)	aboveGroundBiomass	Total dry mass of above-ground seagrass tissue (shoots and leaves) within the sample	g
Supporting variable (biological)	belowGroundBiomass	Total dry mass of below-ground seagrass tissue (roots and rhizomes) within the sample	g
Supporting variable (environmental)	depth	Depth of collection in meters, indexed to some permanent benchmark (e.g., mean lower low water)	m
Supporting variable (environmental)	waterClarity	A measure of how far down light can penetrate through the water column	m
Supporting variable (environmental)	waterTemperature	Temperature of water, typically measured in degrees C, using a thermometer or other sensor	degrees C
Supporting variable (environmental)	salinity	The concentration of dissolved salts in seawater, generally measured in Practical Salinity Units (PSU), a unit based on sea water conductivity, and equivalent to parts per thousand or (o/00) or g/kg.	Practical salinity units (PSU, g salt 1000 g water ⁻¹)
Supporting variable (environmental)	sedimentGrainSize	Distribution of sediment grain sizes within a sample, usually measured by sieving and expressed on the Wentworth or Krumbein phi (φ) scale	Wentworth or Krumbein phi (φ) scale
Supporting variable (other EOVS)	epifaunaAbundance	The number of mobile epifaunal invertebrates sampled from a quantity of seagrass (per seabed area, per g seagrass, or per cm2 seagrass leaf)	number m-2, number g-1 seagrass, or number cm-2 leaf
Supporting variable (other EOVS)	fishAbundance	The number of fishes sampled from a quantity of seagrass (typically per seabed area)	number m ⁻²
Supporting variable (other EOVS)	macroalgaeBiomass	The dry mass of macroalgae sampled from a seabed plot of specified area	g m ⁻²
Supporting variable (other EOVS)	macroalgaeCover	Proportion of substrate in a sample area that is covered (substrate not visible) by macroalgae	Percent
Supporting variable (other EOVS)	epiphyteDryMass	Dry mass of algae sampled from a quantity of seagrass	g g ⁻¹ macrophyte

Tab: Example Event

eventID	eventDate	decimalLatitude	decimalLongitude	featureID	coordinateUncertaintyInMeters	countryCode	datasetName	geobitColumnLabel	institutionCode	institutionID	score	maximumDepthMeters	minimumDepthMeters	parentEventID	samplingProtocol	
MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1	2021-07-22	27.442025	-80.287701	LINESTRING (77.44225 -80.287701, 27.441801 -80.287151)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	1.5	NA		https://doi.org/10.25327/mar.14924214
MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.10	2021-07-22	27.442025	-80.287701	LINESTRING (77.44225 -80.287701, 27.441801 -80.287151)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	1.5	MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.10.sargassum-cover	2021-07-22	27.442025	-80.287701	LINESTRING (77.44225 -80.287701, 27.441801 -80.287151)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	1.5	MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.10	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.10.sargassum-depth	2021-07-22	27.442025	-80.287701	LINESTRING (77.44225 -80.287701, 27.441801 -80.287151)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	1.5	MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.10	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.10.sargassum-ethanol	2021-07-22	27.442025	-80.287701	LINESTRING (77.44225 -80.287701, 27.441801 -80.287151)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	1.5	MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.10	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.8	2021-07-22	27.442025	-80.287701	LINESTRING (77.44225 -80.287701, 27.441801 -80.287151)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	1.5	MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.8	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.8.sargassum-cover	2021-07-22	27.442025	-80.287701	LINESTRING (77.44225 -80.287701, 27.441801 -80.287151)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	1.5	MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.8	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.8.sargassum-ethanol	2021-07-22	27.442025	-80.287701	LINESTRING (77.44225 -80.287701, 27.441801 -80.287151)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	1.5	MarineGEO_USA-RL_sargassum_2021_Beast_Punt_Sargassum_1.8	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1	2021-07-29	27.538845	-80.348370	LINESTRING (77.538842 -80.348371, 27.537048 -80.34848)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	2	NA		https://doi.org/10.25327/mar.14924214
MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.10	2021-07-29	27.538845	-80.348370	LINESTRING (77.538842 -80.348371, 27.537048 -80.34848)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	2	MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.10.sargassum-cover	2021-07-29	27.538845	-80.348370	LINESTRING (77.538842 -80.348371, 27.537048 -80.34848)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	2	MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.10	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.10.sargassum-ethanol	2021-07-29	27.538845	-80.348370	LINESTRING (77.538842 -80.348371, 27.537048 -80.34848)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	2	MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.10	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.8	2021-07-29	27.538845	-80.348370	LINESTRING (77.538842 -80.348371, 27.537048 -80.34848)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	2	MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.8.sargassum-cover	2021-07-29	27.538845	-80.348370	LINESTRING (77.538842 -80.348371, 27.537048 -80.34848)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	2	MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.8	https://doi.org/10.25327/mar.14924214	
MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.8.sargassum-ethanol	2021-07-29	27.538845	-80.348370	LINESTRING (77.538842 -80.348371, 27.537048 -80.34848)	20	US	MarineGEO Global Sargassum Habitat Monitoring Data EPSC-420	sargassum	MarineGEO	MarineGEO	Attribution	3	2	MarineGEO_USA-RL_sargassum_2021_Habur_Branch_Sargassum_1.8	https://doi.org/10.25327/mar.14924214	

Tab: Example Occurrence

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eventID	occurrenceID	basioffRecord	occurrenceStatus	scientificName	scientificNameID	class	family	genus	kingdom	order	phylum	taxonRank	
MarineOEO_USA_IRL_seagrass_2021 Harbor_Branch_Seagrass:1:8 seagrass-cover	MarineOEO_USA_IRL_seagrass_2021 Harbor_Branch_Seagrass:1:8 seagrass-cover208925	HumanObservation	present	Halodule wrightii	um:8d:marine:species.org:taxname:208925	Magnoliopsida	Cymodoceaceae	Halodule	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Harbor_Branch_Seagrass:1:8 seagrass-cover	MarineOEO_USA_IRL_seagrass_2021 Harbor_Branch_Seagrass:1:8 seagrass-cover211788	HumanObservation	present	Acarothophus spicifera	um:8d:marine:species.org:taxname:211788	Funduliphyceae	Rhodomonaceae	Acarothophus	Plantae	Ceramiales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Harbor_Branch_Seagrass:1:16 seagrass-cover	MarineOEO_USA_IRL_seagrass_2021 Harbor_Branch_Seagrass:1:16 seagrass-cover374722	HumanObservation	present	Halophila engelmannii	um:8d:marine:species.org:taxname:374722	Magnoliopsida	Hydrocharitaceae	Halophila	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-cover	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-cover208925	HumanObservation	present	Halodule wrightii	um:8d:marine:species.org:taxname:208925	Magnoliopsida	Cymodoceaceae	Halodule	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora22486	HumanObservation	present	Nereididae	um:8d:marine:species.org:taxname:22486	Polychaeta	Nereididae	NA	Animalia	Phyllodocera	NA	family	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora168361	HumanObservation	present	Hyporhamphus aeneolatus	um:8d:marine:species.org:taxname:168361	Malaconostoma	Hyporhamphidae	Hyporhamphus	Animalia	Amphipoda	Amnelliella	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora1268	HumanObservation	present	Sipuncula	um:8d:marine:species.org:taxname:1268	NA	NA	Animalia	Amphipoda	Amnelliella	order		
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora87761	HumanObservation	present	Phoronis viles	um:8d:marine:species.org:taxname:87761	Gastropoda	Nassariidae	Phoronis	Animalia	Neogastropoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora1135	HumanObservation	present	Amphipoda	um:8d:marine:species.org:taxname:1135	Malaconostoma	NA	Animalia	Amphipoda	Amnelliella	order		
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora102401	HumanObservation	present	Erithonius brasiliensis	um:8d:marine:species.org:taxname:102401	Malaconostoma	Ischyronidae	Erithonius	Animalia	Amphipoda	Amnelliella	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora421588	HumanObservation	present	Carapax curdipi	um:8d:marine:species.org:taxname:421588	Malaconostoma	Ischyronidae	Carapax	Animalia	Amphipoda	Amnelliella	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora160102	HumanObservation	present	Astyris lunata	um:8d:marine:species.org:taxname:160102	Gastropoda	Columbellidae	Astyris	Animalia	Neogastropoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora160174	HumanObservation	present	Bitellum varium	um:8d:marine:species.org:taxname:160174	Gastropoda	Cerithiidae	Bitellum	Animalia	Caenogastropoda	Incertae sedis	Mollusca	species
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora160065	HumanObservation	present	Acteocina canaliculata	um:8d:marine:species.org:taxname:160065	Gastropoda	Tornatellidae	Acteocina	Animalia	Cephalopoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora426149	HumanObservation	present	Prunum apiculatum	um:8d:marine:species.org:taxname:426149	Gastropoda	Margarinellidae	Prunum	Animalia	Neogastropoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora160062	HumanObservation	present	Cymatulus constrictus	um:8d:marine:species.org:taxname:160062	Malaconostoma	Amphipoda	Cymatulus	Animalia	Amphipoda	Amnelliella	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora419588	HumanObservation	present	Caecum fontanum	um:8d:marine:species.org:taxname:419588	Gastropoda	Caecidae	Caecum	Animalia	Littorinophora	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora160228	HumanObservation	present	Crepidula conveasa	um:8d:marine:species.org:taxname:160228	Gastropoda	Calyptraeidae	Crepidula	Animalia	Littorinophora	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora87761	HumanObservation	present	Phoronis viles	um:8d:marine:species.org:taxname:87761	Gastropoda	Nassariidae	Phoronis	Animalia	Neogastropoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora1268	HumanObservation	present	Sipuncula	um:8d:marine:species.org:taxname:1268	NA	NA	Animalia	Amphipoda	Amnelliella	order		
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora138040	HumanObservation	present	Bulla nitida	um:8d:marine:species.org:taxname:138040	Gastropoda	Bullidae	Bulla	Animalia	Cephalopoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora514508	HumanObservation	present	Cuscuta americana	um:8d:marine:species.org:taxname:514508	Malaconostoma	Palaeomonidae	Cuscuta	Animalia	Decapoda	Amphipoda	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora160065	HumanObservation	present	Acteocina canaliculata	um:8d:marine:species.org:taxname:160065	Gastropoda	Tornatellidae	Acteocina	Animalia	Cephalopoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora1065	HumanObservation	present	Amphipoda	um:8d:marine:species.org:taxname:1065	NA	NA	Animalia	NA	Amphipoda	phylum		
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora261817	HumanObservation	present	Paracerasia caudata	um:8d:marine:species.org:taxname:261817	Malaconostoma	Sphaerostomatidae	Paracerasia	Animalia	Isopoda	Amphipoda	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora160102	HumanObservation	present	Astyris lunata	um:8d:marine:species.org:taxname:160102	Gastropoda	Columbellidae	Astyris	Animalia	Neogastropoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora160174	HumanObservation	present	Bitellum varium	um:8d:marine:species.org:taxname:160174	Gastropoda	Cerithiidae	Bitellum	Animalia	Caenogastropoda	Incertae sedis	Mollusca	species
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora421588	HumanObservation	present	Carapax curdipi	um:8d:marine:species.org:taxname:421588	Malaconostoma	Ischyronidae	Carapax	Animalia	Amphipoda	Amnelliella	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora419588	HumanObservation	present	Caecum fontanum	um:8d:marine:species.org:taxname:419588	Gastropoda	Caecidae	Caecum	Animalia	Littorinophora	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora155092	HumanObservation	present	Cymatulus constrictus	um:8d:marine:species.org:taxname:155092	Malaconostoma	Amphipoda	Cymatulus	Animalia	Amphipoda	Amnelliella	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora1135	HumanObservation	present	Amphipoda	um:8d:marine:species.org:taxname:1135	Malaconostoma	NA	Animalia	Amphipoda	Amnelliella	order		
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora102401	HumanObservation	present	Erithonius brasiliensis	um:8d:marine:species.org:taxname:102401	Malaconostoma	Ischyronidae	Erithonius	Animalia	Amphipoda	Amnelliella	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora420149	HumanObservation	present	Prunum apiculatum	um:8d:marine:species.org:taxname:420149	Gastropoda	Margarinellidae	Prunum	Animalia	Neogastropoda	Mollusca	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora101	HumanObservation	present	Gastropoda	um:8d:marine:species.org:taxname:101	Gastropoda	NA	Animalia	NA	Amphipoda	class		
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora22496	HumanObservation	present	Nereididae	um:8d:marine:species.org:taxname:22496	Polychaeta	Nereididae	NA	Animalia	Phyllodocera	NA	family	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora247338	HumanObservation	present	Hegania ripes	um:8d:marine:species.org:taxname:247338	Malaconostoma	Leptochelidae	Hegania	Animalia	Tanaidacea	Amphipoda	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora160228	HumanObservation	present	Halodule wrightii	um:8d:marine:species.org:taxname:160228	Magnoliopsida	Cymodoceaceae	Halodule	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora208925	HumanObservation	present	Syringodium filiforme	um:8d:marine:species.org:taxname:208925	Magnoliopsida	Cymodoceaceae	Syringodium	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora217165	HumanObservation	present	Acarothophus spicifera	um:8d:marine:species.org:taxname:217165	Funduliphyceae	Rhodomonaceae	Acarothophus	Plantae	Ceramiales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora211788	HumanObservation	present	Halophila engelmannii	um:8d:marine:species.org:taxname:211788	Funduliphyceae	Rhodomonaceae	Halophila	Plantae	Ceramiales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-epiflora145713	HumanObservation	present	Halophila decipiens	um:8d:marine:species.org:taxname:145713	Magnoliopsida	Hydrocharitaceae	Halophila	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora145836	HumanObservation	present	Hypnea spiralis	um:8d:marine:species.org:taxname:145836	Funduliphyceae	Cystodermaceae	Hypnea	Plantae	Gigartinales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-epiflora146036	HumanObservation	present	Halophila spicifera	um:8d:marine:species.org:taxname:146036	Funduliphyceae	Rhodomonaceae	Halophila	Plantae	Ceramiales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Harbor_Branch_Seagrass:1:8 seagrass-shoots	MarineOEO_USA_IRL_seagrass_2021 Harbor_Branch_Seagrass:1:8 seagrass-shoots208925	HumanObservation	present	Halodule wrightii	um:8d:marine:species.org:taxname:208925	Magnoliopsida	Cymodoceaceae	Halodule	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-shoots	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:16 seagrass-shoots208925	HumanObservation	present	Syringodium filiforme	um:8d:marine:species.org:taxname:208925	Magnoliopsida	Cymodoceaceae	Syringodium	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-shoots	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-shoots208925	HumanObservation	present	Halodule wrightii	um:8d:marine:species.org:taxname:208925	Magnoliopsida	Cymodoceaceae	Halodule	Plantae	Alismatales	NA	species	
MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-shoots	MarineOEO_USA_IRL_seagrass_2021 Bear_Point_Seagrass:1:8 seagrass-shoots208925	HumanObservation	present	Halodule wrightii	um:8d:marine:species.org:taxname:208925	Magnoliopsida	Cymodoceaceae	Halodule	Plantae	Alismatales	NA	species	

extend() occurrence()

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